

Ch 3 – Factors of Polynomials

Remainder Theorem:

When the polynomial $f(x)$ is divided by $(x - a)$, the remainder is $f(a)$.

Factor Theorem:

If $f(a) = 0$, $(x - a)$ is a factor of $f(x)$

Factoring Polynomials:

Eg: $3x + 81$

$$3(x + 27)$$

$$3x + 81 = 3(x + 27)$$

Eg: $3x^2 + 17x + 10$

$$x^2 + 17x + 30$$

$$(x + 15)(x + 2)$$

$$(x + 5)(3x + 2)$$

$$3x^2 + 17x + 10 = (x + 5)(3x + 2)$$

Eg: $x^3 + 10x^2 + 27x + 18$

Factors of 18 = $\pm 1, \pm 2, \pm 3, \pm 6, \pm 9$

$$(1)^3 + 10(1)^2 + 27(1) + 18 = 1 + 10 + 27 + 18 = 56$$

$56 \neq 0$ so, 1 is not a factor of the polynomial

$$(-1)^3 + 10(-1)^2 + 27(-1) + 18 = -1 + 10 - 27 + 18 = 0$$

$0 = 0$ so, -1 is a factor of the polynomial

$$\begin{array}{r|rrrr}
 x = -1 & 1 & 10 & 27 & 18 \\
 & & -1 & -9 & -18 \\
 \hline
 & 1 & 9 & 18 & 0
 \end{array}$$

$$x^3 + 10x^2 + 27x + 18 = (x + 1)(x^2 + 9x + 18)$$

$$x^2 + 9x + 18 = (x + 3)(x + 6)$$

$$x^3 + 10x^2 + 27x + 18 = (x + 1)(x + 3)(x + 6)$$

Solving Cubic Equations:

- 1) Ensure equation = 0
- 2) Factor the equation
- 3) Equate to 0
- 4) Find x values